

Practical course

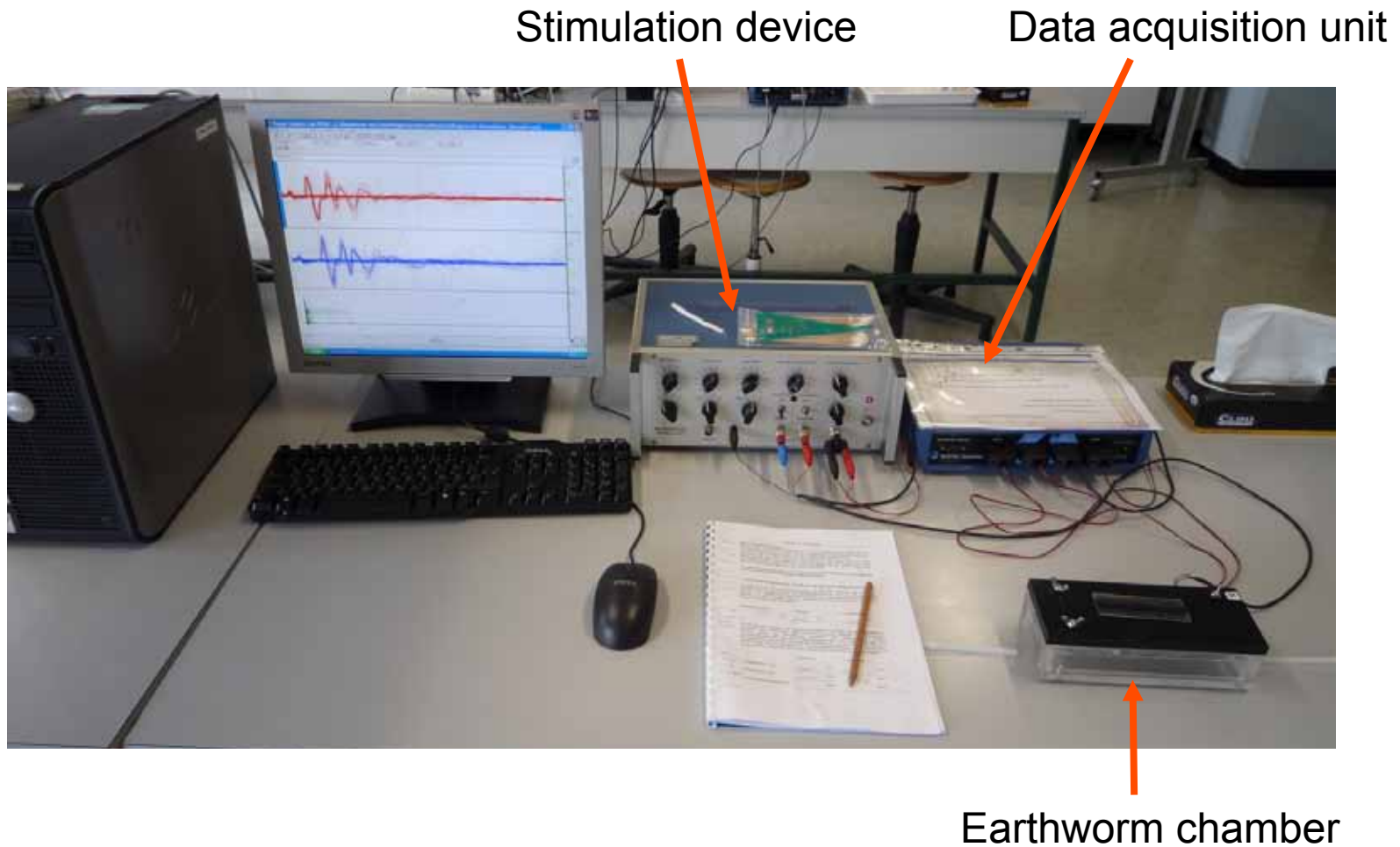
Nerve fiber

Action potential recording
from the median giant fiber
of the intact earthworm



Institute of Cellular and Integrative Physiology,
University of Hamburg, Germany

Experimental setup

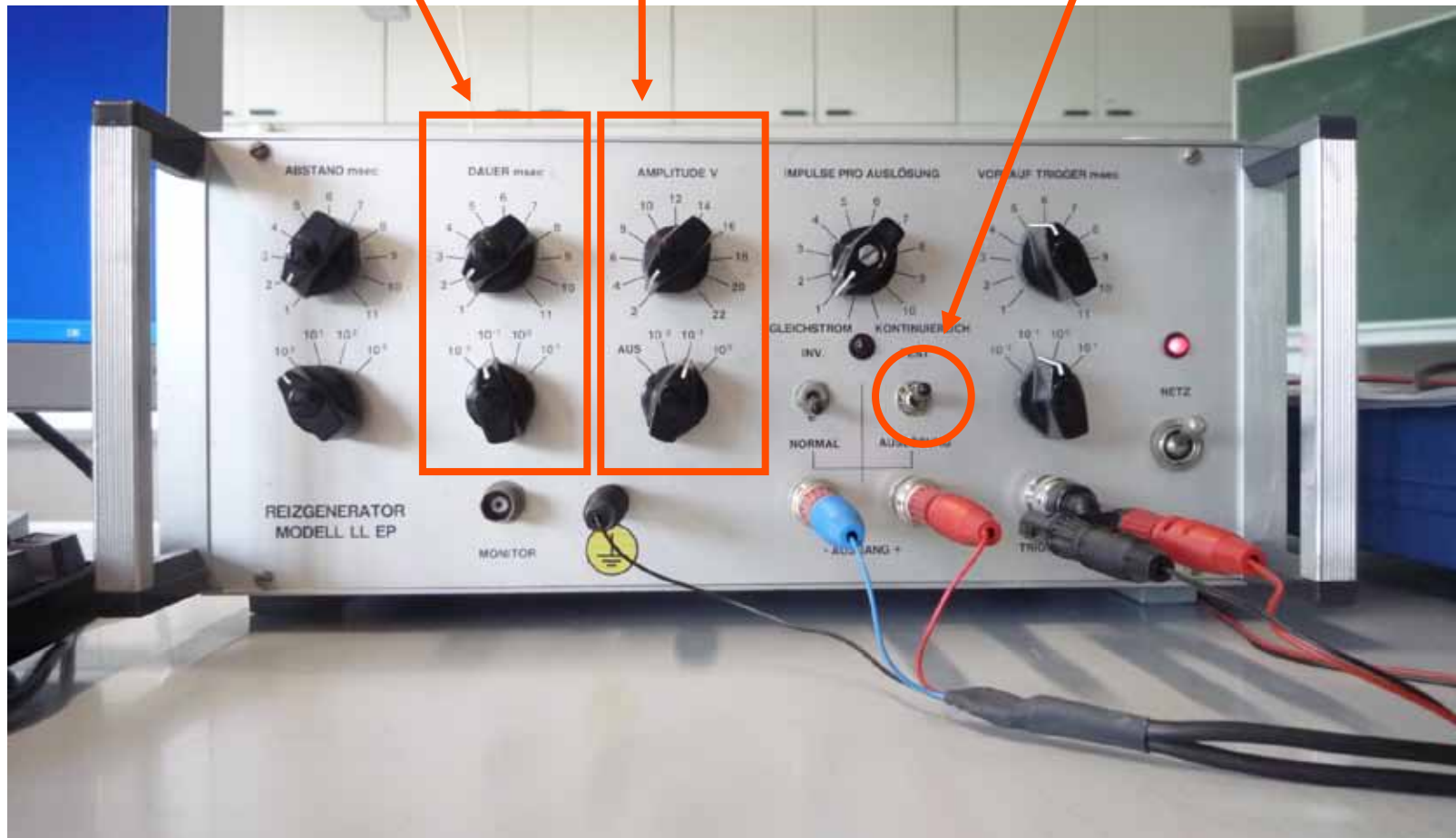


Use (only!) these buttons:

Stimulus
duration*

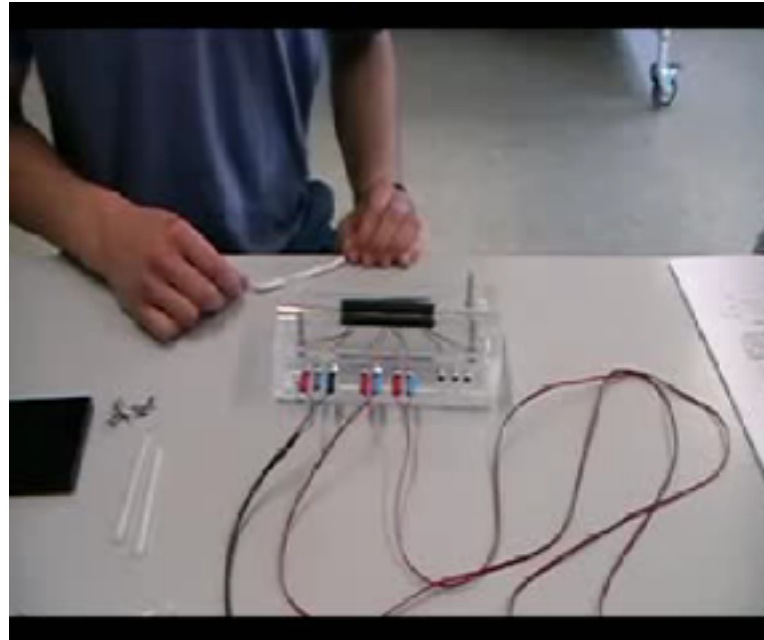
Stimulus
amplitude*

Stimulus trigger



* Initial settings used for the experiments: 0.1 ms, 0.5 V

Practicing the use of the computer program and the operation of the stimulation device with the “worm dummy”



(Movie: Worm dummy)

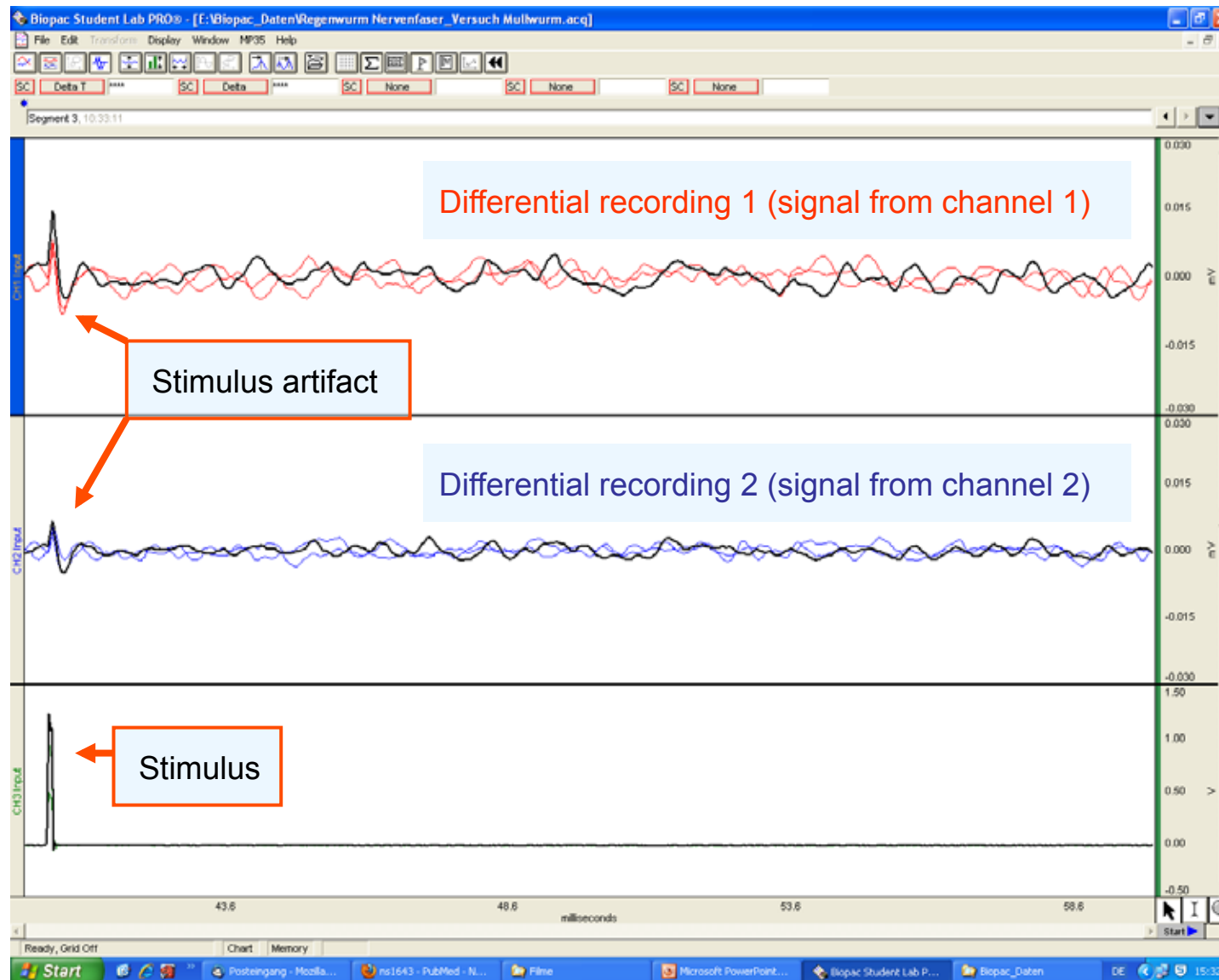
Practicing the use of the computer program and the operation of the stimulation device with the “worm dummy”

Open the BIOPAC file “Worm dummy”

To get familiar with the experimental setup,

1. change stimulus duration and/or stimulus amplitude (stimulation device)
2. click “start” (computer program)
3. trigger the stimulus (stimulation device)
4. inspect the recording
5. continue with step 1

Practicing with the “worm dummy” –
the recordings should look like this:



After a successful dry run: Fetch the earthworms



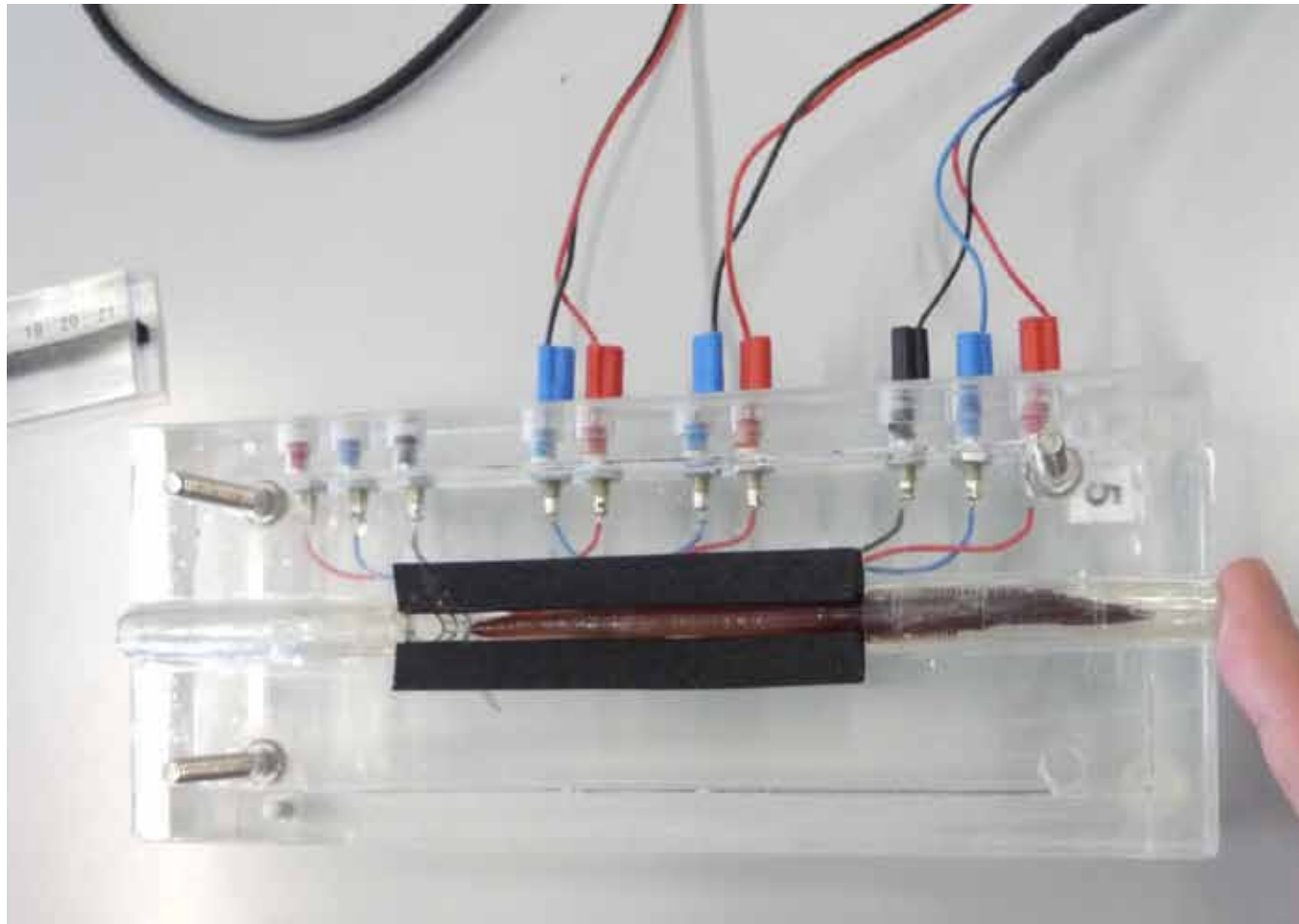
(Movie: fetch worms)

After the practical course, the earthworms will be set free in an appropriate environment – they will **not** be used as bait for fishing!

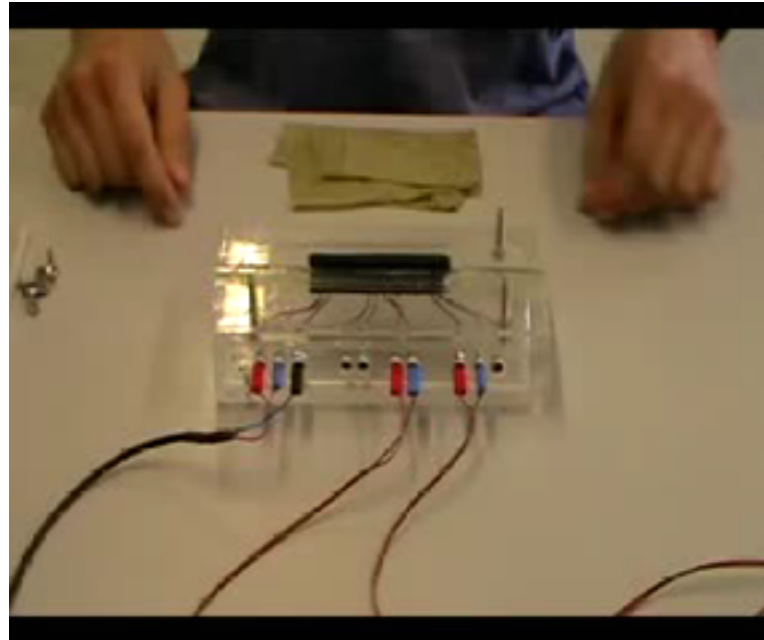
Earthworm after rinsing



How does the earthworm get into the chamber?

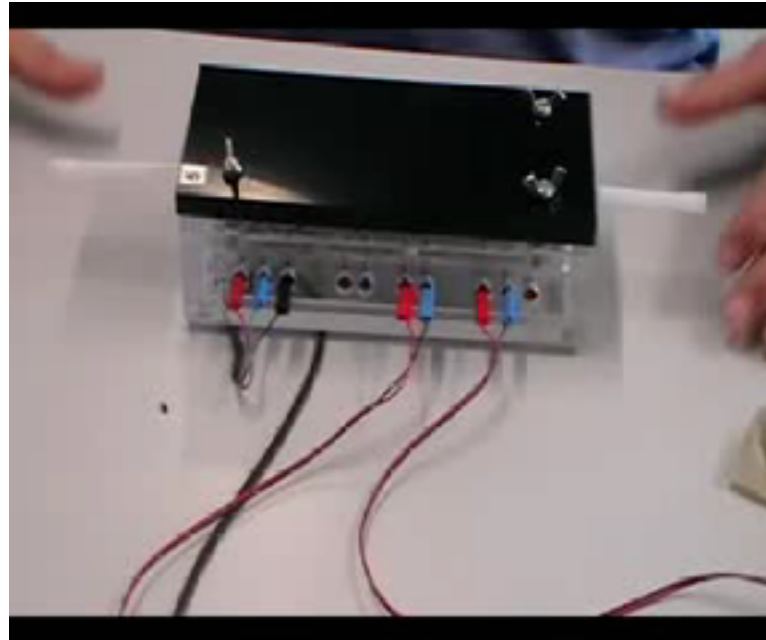


It enters the chamber on its own completely unforced:



(Movie: worm into chamber)

Please check:
Does the earthworm have proper contact
with the silver electrodes?



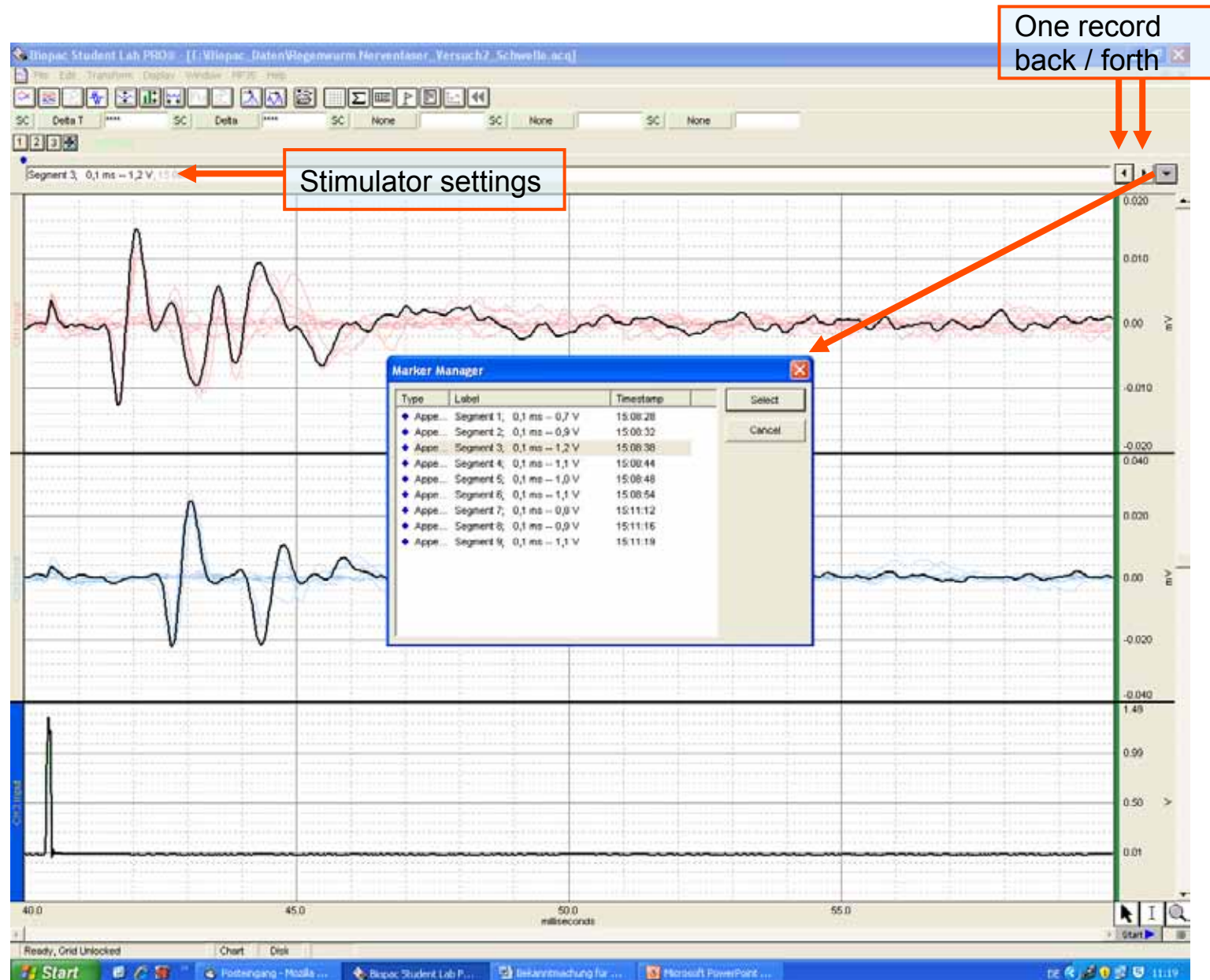
(Movie: turn chamber)

Finding the stimulus threshold



(Movie: find threshold)

Stimulus threshold – the recordings should look like this:

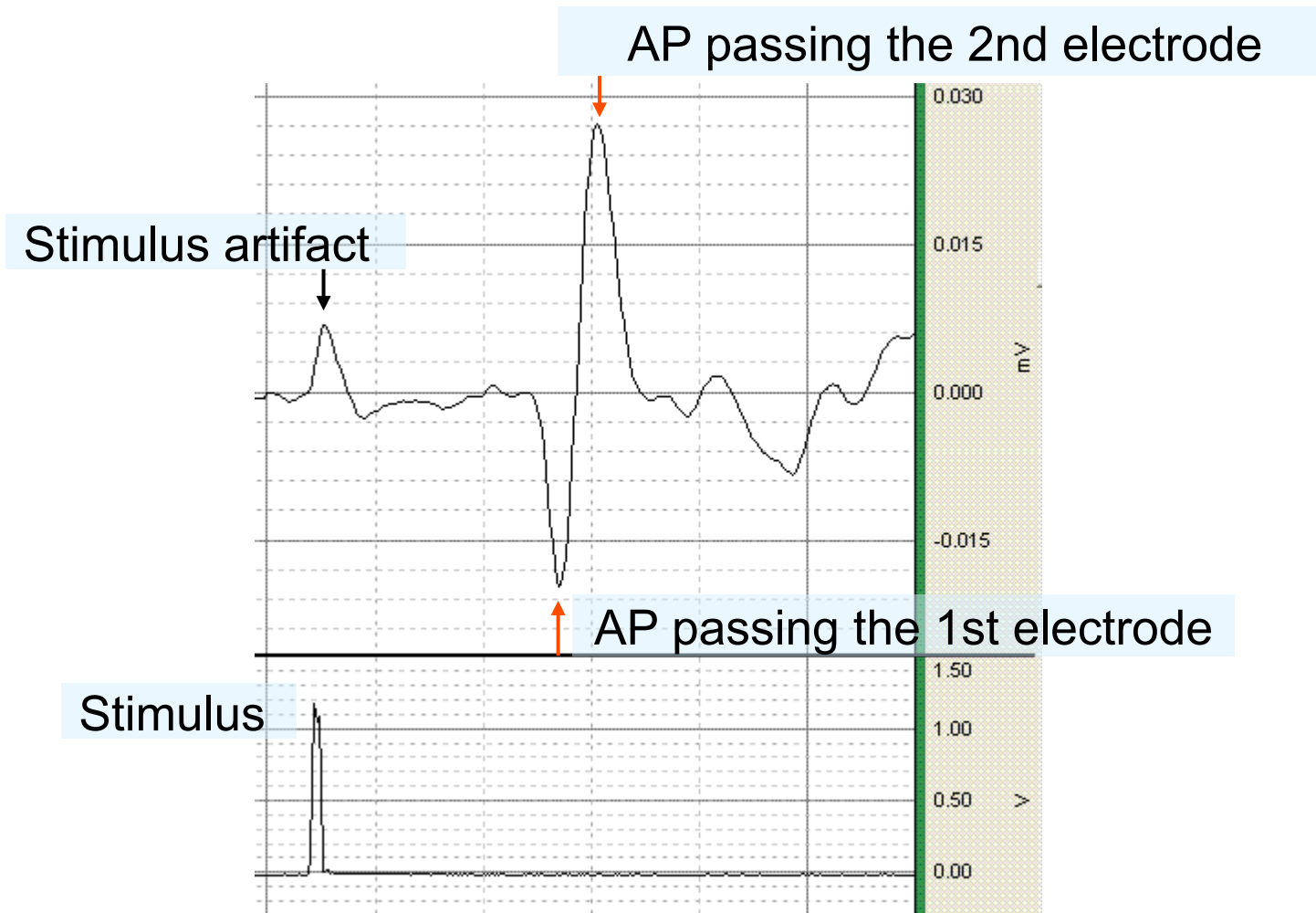


Finding the stimulus threshold

Open the BIOPAC file “Threshold”

Extracellular differential recording: Why do we get a biphasic AP?

*Measurement of the **potential difference** between a pair of electrodes*

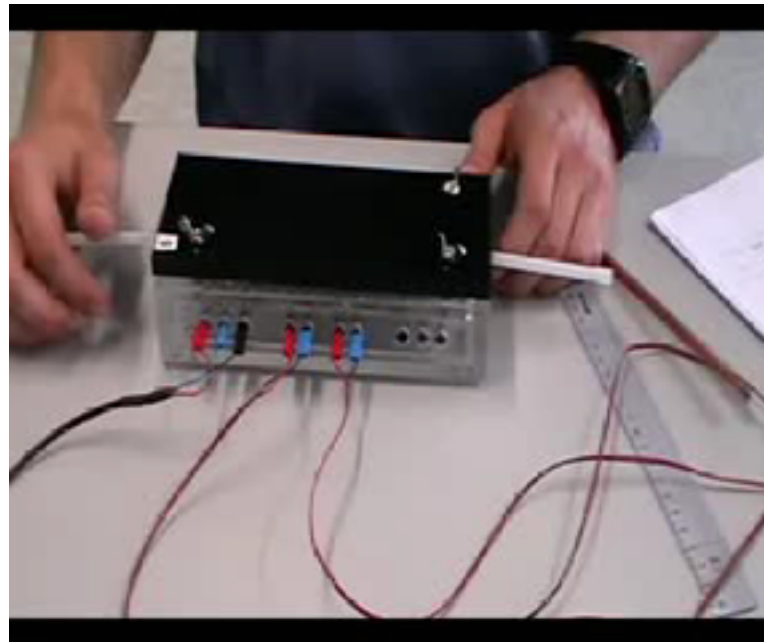


Determining the AP conduction velocity

Different electrode positions in 2 consecutive trials:

Recording Channel 1: E4 – E5 (constant)

Recording Channel 2: first E6 – E7, then E8 – E9



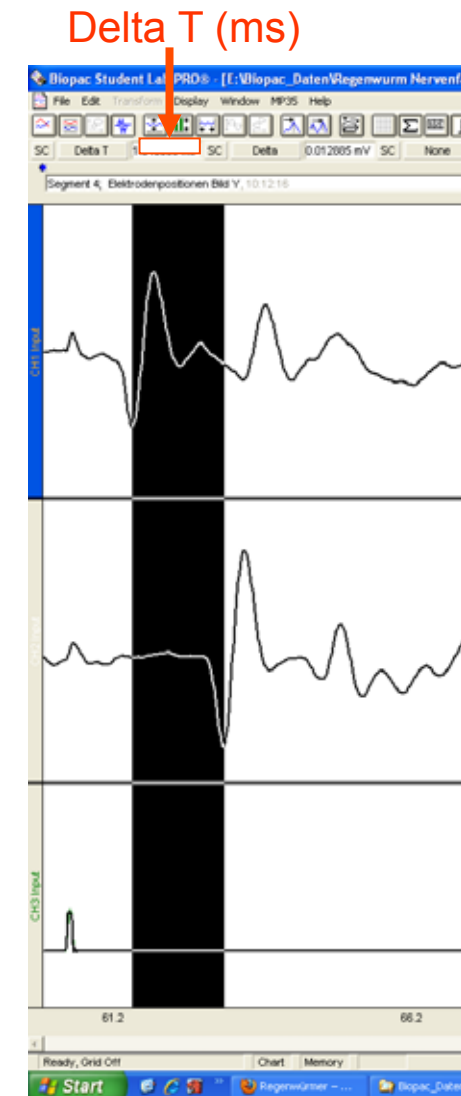
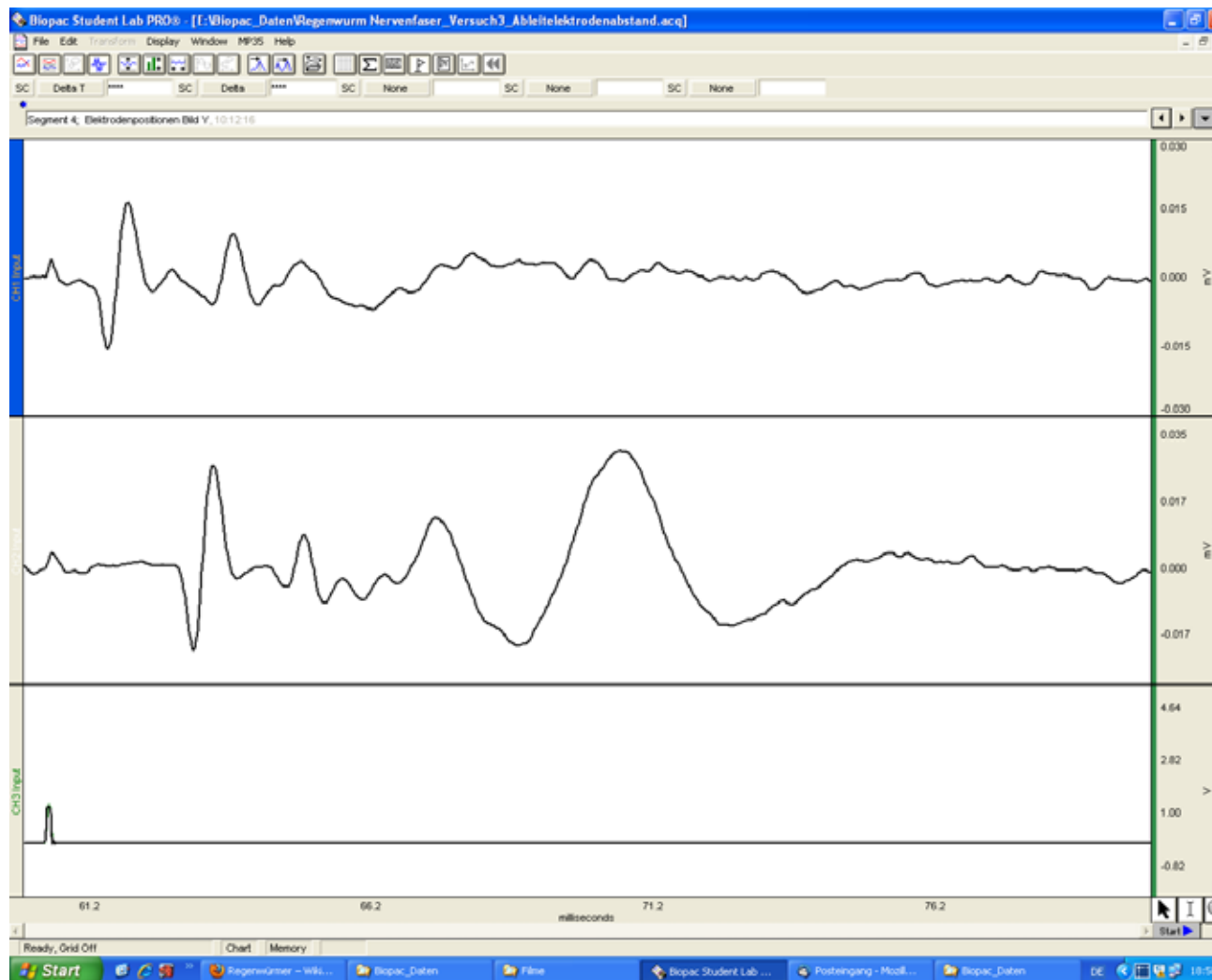
(Movie: change electrode position)

Determining the AP conduction velocity

Open BIOPAC file “Conduction velocity”
and analyze the data provided

Determining the AP conduction velocity

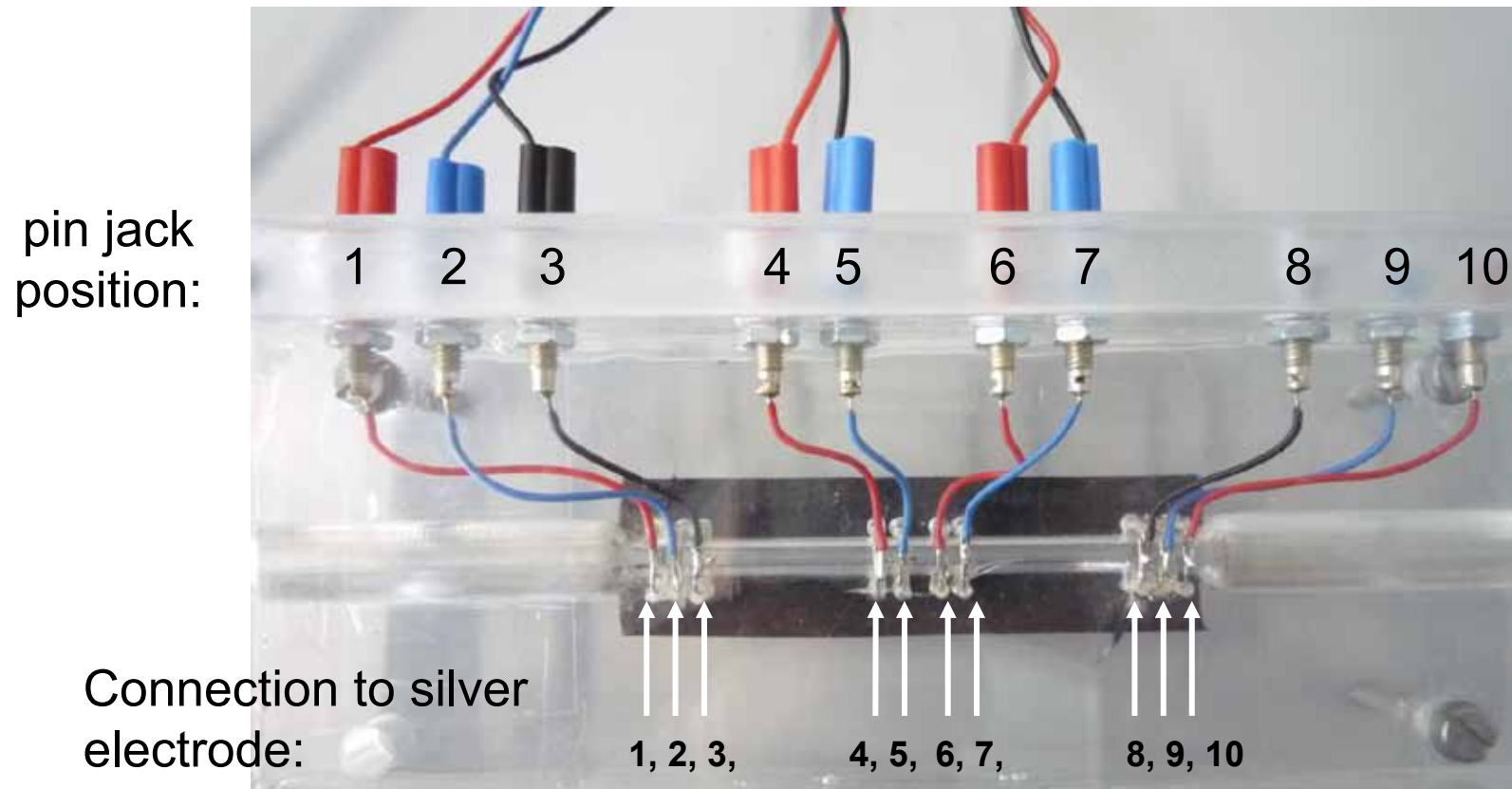
A) Measure the time difference between corresponding parts of the biphasic AP from the two differential recordings



Determining the AP conduction velocity

B) Measure the distance between the corresponding electrodes of the two differential recordings

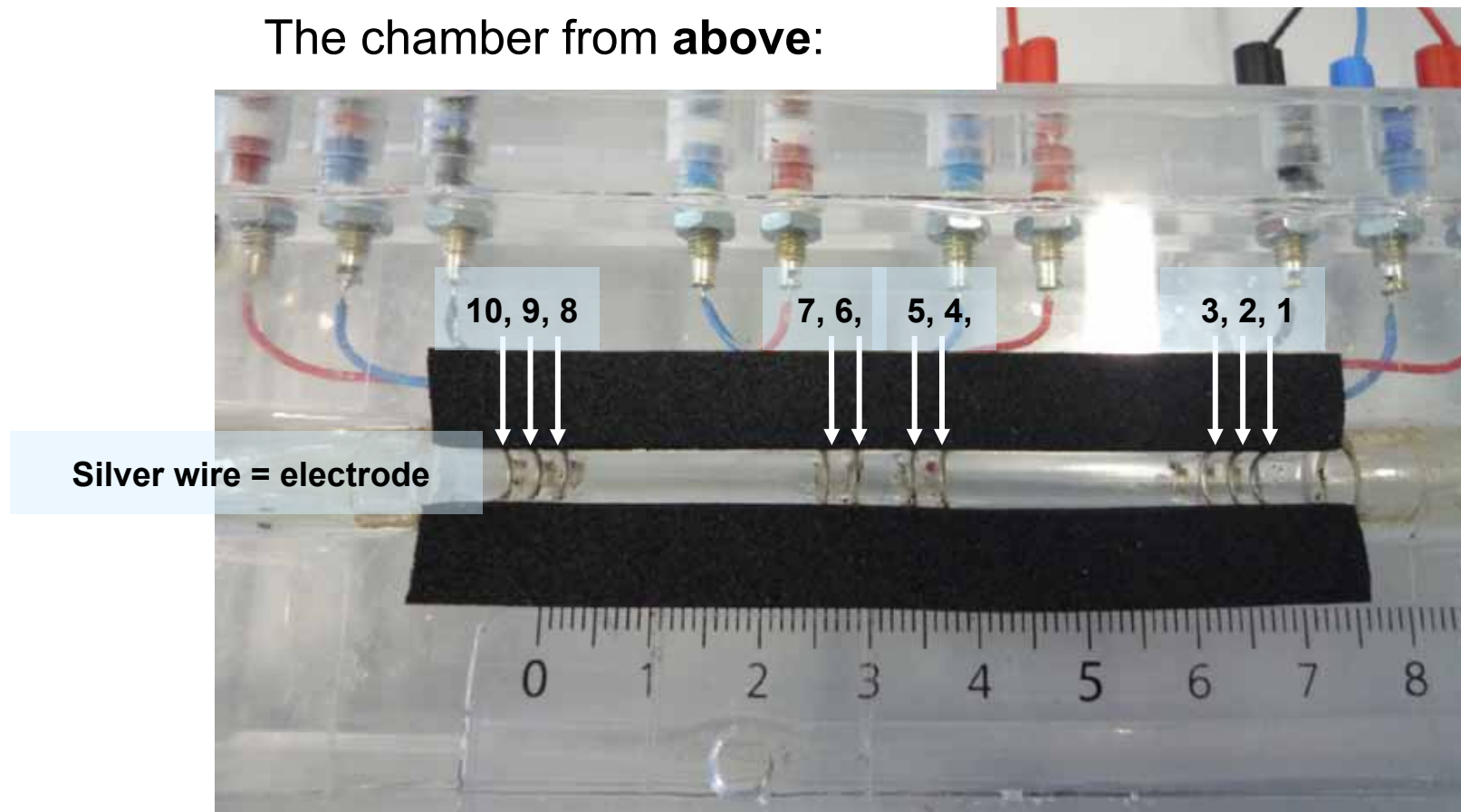
How the earthworm chamber looks from the **bottom**:



Determining the AP conduction velocity

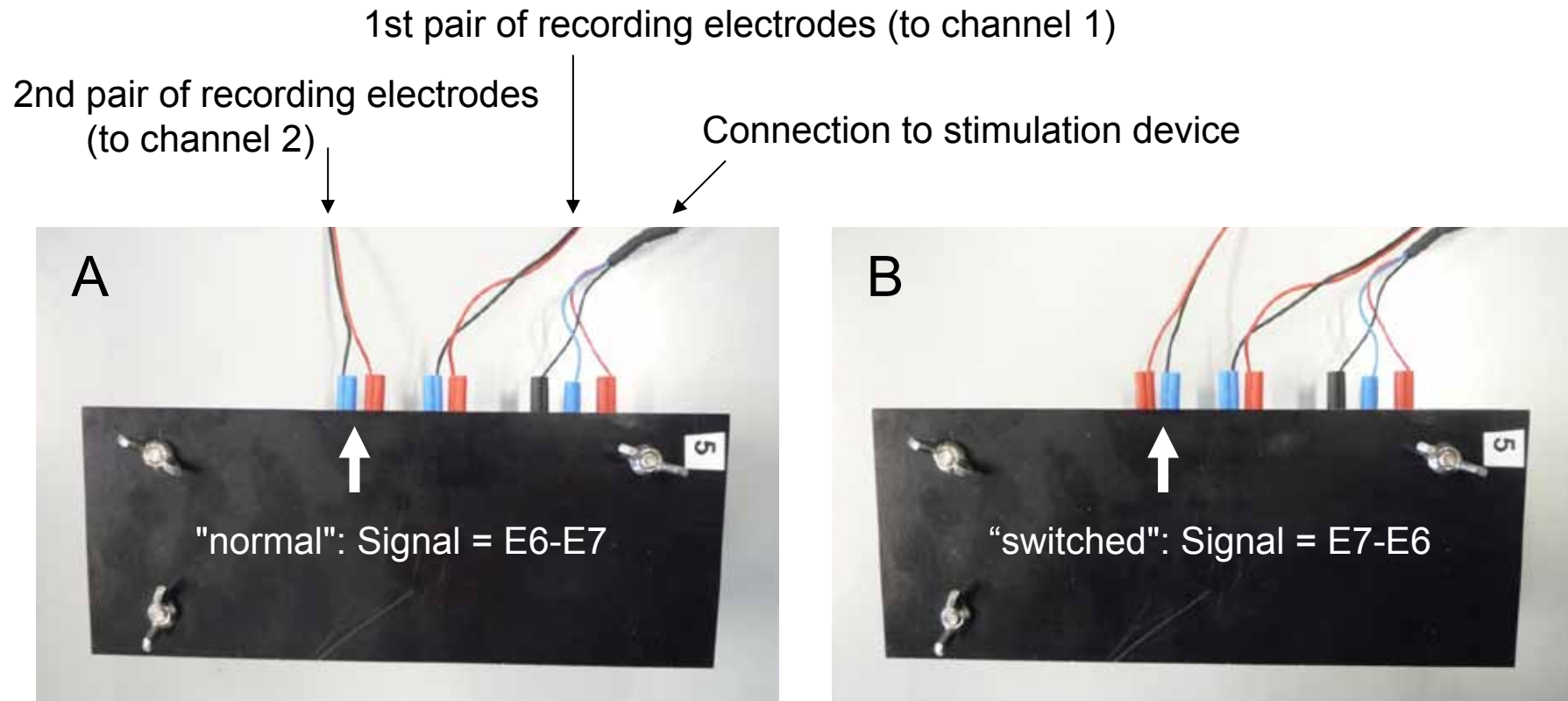
B) Measure the distance between the corresponding electrodes of the two differential recordings

The chamber from **above**:



Switching the sequence of the 2nd pair of recording electrodes

You always need a pair of electrodes to get a differential recording



The operation amplifier "calculates" the **difference** in the potentials of a pair of electrodes, connected via the red and blue plugs ("red" – "blue")

Switching the sequence of the 2nd pair of recording electrodes

Open the BIOPAC file: “Switch electrodes”

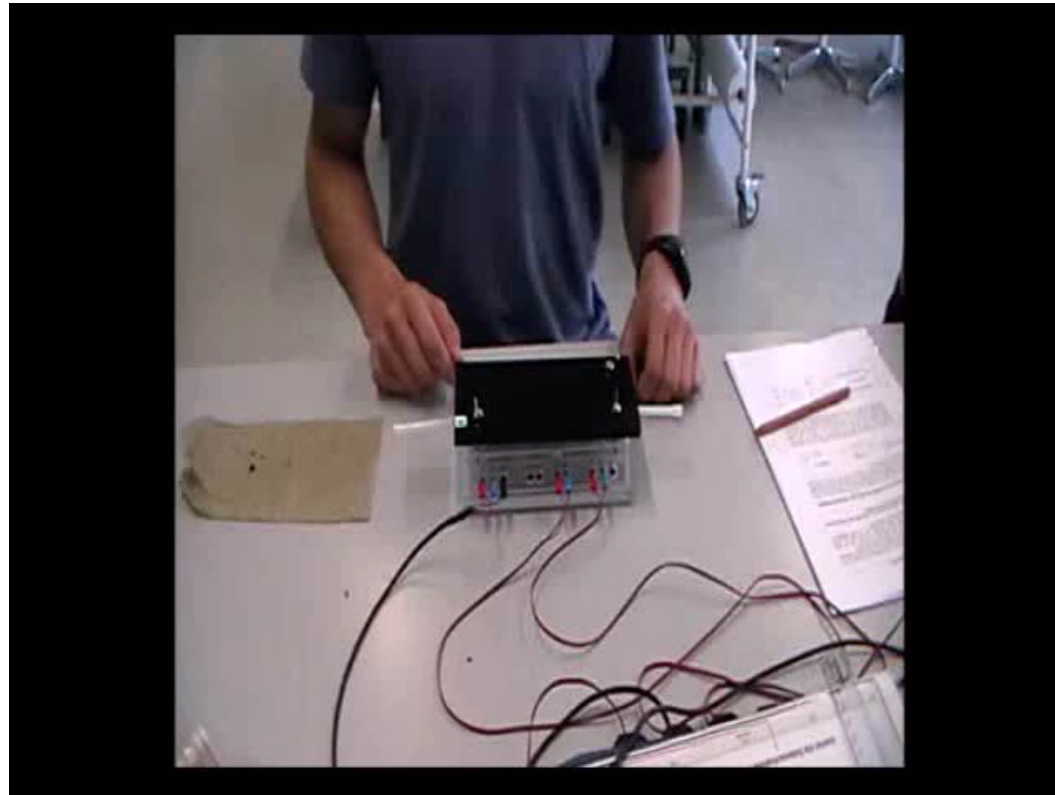
Describe the effect of switching the connections!

Experiment Rheobase - Chronaxie

Open the BIOPAC file: "Chronaxie"

1. List the threshold amplitudes for all tested stimulus durations;
2. Create the strength-duration plot and graphically determine rheobase and chronaxie.

How does the worm get out of the chamber?



(Movie: worm out of chamber)

Do not forget to clean the chamber!



(Movie: clean chamber)

Thanks to...

...as well as
Peter, Charlotte and Jonas

